

Enhanced RNA secondary structure prediction through integrative deep learning and structural context analysis

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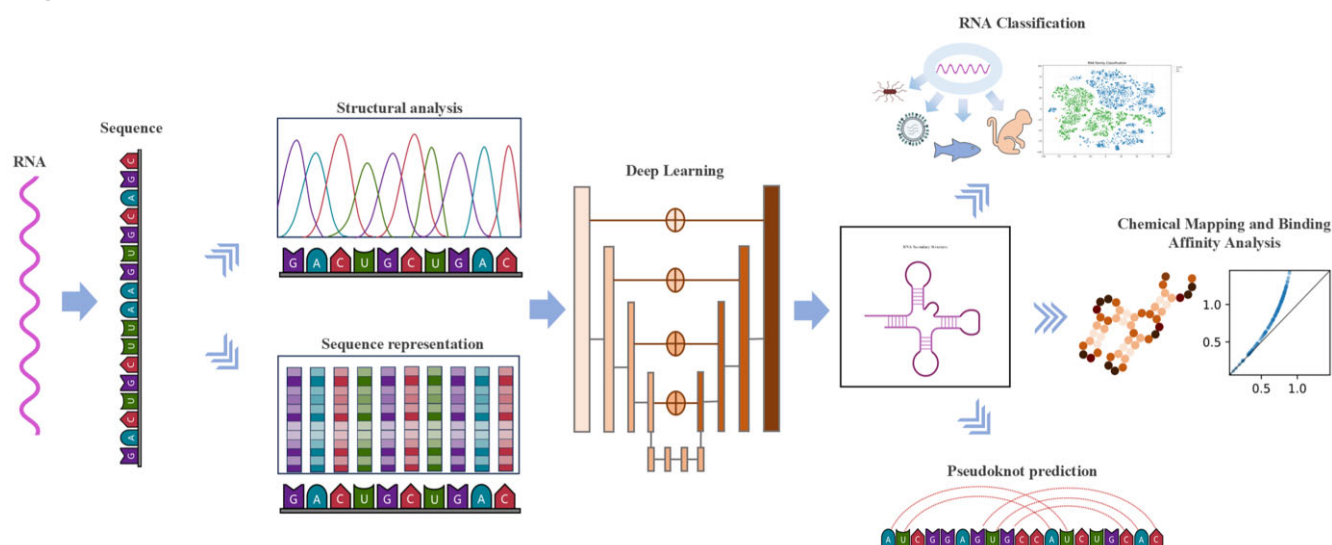
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Abstract

Analyzing RNA secondary structures plays a crucial role in elucidating the functional mechanisms of RNA. Despite advances in RNA structure determination, these methods are low throughput and resource-intensive. While machine learning-based models have achieved remarkable performance in terms of prediction accuracy, challenges such as data scarcity and overfitting remain common. Here, we introduce a phased learning strategy that integrates RNA sequence and structural context information to mitigate the risk of overfitting and employs pairing constraints to train the model on folding scores. This approach effectively addresses both local and long-range nucleotide interactions, substantially improving the robustness of RNA secondary structure predictions. Our comprehensive analysis across multiple benchmarking datasets demonstrated that the performance of our model (DSRNAFold) was superior to that of existing methods, especially in pseudoknot recognition and chemical mapping activity prediction, where our approach showed positive performance.

Graphical abstract



Introduction

RNA molecules serve many roles, reflecting their importance in cellular and molecular biology [1–3]. Apart from its role as a messenger between DNA and proteins, RNA can act as a catalyst, a regulator of gene expression, and an essen-

tial component of cellular machinery, such as ribosomes and spliceosomes [4]. The functional diversity of RNA can be attributed to its complex structure, with a hierarchical organization ranging from a linear sequence of nucleotides to intricate 3D arrangements [5, 6]. In this hierarchy, the secondary

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structure of RNA, characterized by specific interactions between nucleotides, such as canonical and noncanonical base pairing, is of critical importance [7]. This structure gives rise to an array of motifs, including but not limited to hairpin loops, bulges, and internal loops, each with unique biological functions [8]. The secondary structure thus not only determines the three-dimensional architecture of RNA but also underpins its cellular roles, including facilitating the formation of the RNA-induced silencing complex and catalysis in ribosomes and acting as a substrate in viral replication mechanisms [9].

The gold standard for determining the structure of RNA has traditionally been experimental techniques such as X-ray crystallography, nuclear magnetic resonance spectroscopy, and cryo-electron microscopy [10]. While high structural resolution can be achieved using these methods, there are many challenges. For example, these methods are resource intensive, in terms of both the specialized skills required for data acquisition and the financial costs associated with maintaining complex instrumentation [11]. Furthermore, these techniques are characterized by their low throughput, thereby resulting in a limited subset of RNAs for which structures have been experimentally determined. To mitigate these limitations, emerging experimental methodologies have been developed to infer nucleotide base pairing through the use of enzymatic, chemical, and crosslinking probes in combination with high-throughput sequencing techniques [12–14]. While these approaches hold promise, they remain in the nascent stages of development and are currently incapable of achieving base pair resolution at the single-nucleotide level.

Against this backdrop, computational methods have emerged as a more scalable and convenient alternative for RNA secondary structure prediction. Since the advent of computational techniques in the 1970s, various prediction methodologies have been developed. The majority of these approaches aim to determine a structure characterized by the minimum free energy (MFE), aligning with the prevailing hypothesis that RNA molecules are likely to adopt an MFE conformation analogous to that of proteins [15]. The free energy of each nearest-neighbor loop can be calculated by summing the free energy parameters that characterize the loop. The free energy parameters are determined in advance by methods such as optical melting [16]. The free energy of an entire RNA secondary structure is calculated by summing the free energy of the decomposed nearest-neighbor loops. Computational methodologies for RNA secondary structure prediction predominantly utilize algorithms rooted in dynamic programming, such as the widely recognized Zuker algorithm [17]. These algorithms are often guided by thermodynamic models to determine the most energetically stable conformation. Over the years, many software applications have been developed to integrate these predictive frameworks [18–20]. Nevertheless, over the past decade, there has been a noticeable stagnation in both the accuracy and computational speed of these approaches [6]. One significant limitation of these approaches is their inability to accurately model nonnested patterns of base pairs, known as pseudoknots. Such patterns involve intricate arrangements, including two stem–loop structures, where one stem intercalates between the halves of another. The prediction of secondary structures with pseudoknots has proven to be computationally challenging and is categorized as an NP-complete problem within the energy minimization framework [21]. Moreover, the computational burden associated with these algorithms becomes increasingly onerous when they are

applied to extensive or multifaceted datasets. Thus, these algorithms are suboptimal in genome-scale analyses or scenarios where expedient data interpretation is imperative.

In recent years, with the development of deep learning in bioinformatics [22, 23], the field of RNA secondary structure prediction has transformed, and novel and innovative approaches through which RNA structures can be analyzed and predicted have been introduced. Pioneering methods, such as SPOT-RNA [24] and E2Efold [25], leverage deep neural networks and formulate the prediction of RNA secondary structures as binary classification problems and employ sophisticated architectures such as long short-term memory and transformer encoders to automatically capture long-range nucleotide interactions. However, despite their innovative approaches, these methods face challenges related to extensive parameterization, which can lead to computational inefficiency and a susceptibility to overfitting [26]. Diverse methods aimed at mitigating these challenges have been introduced. For instance, Ufold [27], a novel approach, treats RNA sequences as 2D “images” and employs an encoder–decoder framework to extract interaction features of input sequences. Conversely, MXfold2 [28] combines deep learning with existing thermodynamic knowledge, employing a deep neural network trained with a max-margin framework and thermodynamic regularization, ensuring a delicate balance between predictive accuracy and computational robustness. Additional methods such as LinearFold [29] and CentroidFold [30] aim to enhance the speed of dynamic programming, while others, such as CONTRAfold [31] and ContextFold [32], leverage machine learning techniques to train scoring parameters for decomposed substructures from reference structures, offering alternative perspectives in RNA secondary structure prediction. Despite these advancements, researchers continue to navigate through a myriad of challenges, particularly pertaining to overfitting, computational efficiency, and ensuring broad applicability and accuracy across various RNA structures [33, 34]. As the field continues to evolve, the development of models that are not only accurate but also computationally efficient and capable of generalizing across a wide array of RNA structures, including those yet to be discovered, will play a pivotal role in advancing RNA secondary structure prediction.

In the field of RNA secondary structure prediction, the incorporation of transfer learning into deep neural networks has emerged as a promising strategy to mitigate challenges associated with model training and data requirements [24]. Exploiting the potential of transfer learning, a technique that applies knowledge from a source task to enhance performance on a related target task, has become a pivotal strategy in RNA secondary structure prediction, especially when dealing with scarce labeled data. The complexity and diversity of RNA structures, coupled with often limited labeled data for specific sequences, present a formidable challenge for training models from scratch. Furthermore, latent generalized features and patterns within RNA structures that are common across various RNA types can be discovered and utilized, thereby providing a robust foundation for fine-tuning models for specific prediction tasks. Utilizing transfer learning in RNA secondary structure prediction holds promise in terms of potentially reducing computational demands and revealing novel insights into RNA structuring by leveraging preexisting models and knowledge.

Building upon this promising strategy, the present study introduces a deep learning-centric approach for predicting RNA

secondary structures, integrating sequence and structural information within a uniquely formulated architecture. This architecture is specifically constructed to autonomously navigate both local and long-range nucleotide interactions, circumvent the reliance on predetermined thermodynamic parameters, and enable the capability to autonomously decipher complex structural motifs from data, not only underscoring the model's computational efficiency but also positioning it as a potent tool for exhaustive RNA structure–function studies. This initiative approach, which represents a pivotal advancement in RNA biology, has applications in conducting fundamental biological research, conceptualizing targeted therapeutics, and propelling personalized medicine.

In this paper, we employ a phased learning strategy and propose a deep learning-centric approach for the prediction of RNA secondary structures. By leveraging a deep learning network, we transform the position of each nucleotide in the RNA sequence into an intricate representation that captures contextual features of the sequence. Simultaneously, this study introduces a method to analyze RNA structures based on base pairing weights and local context effects, with a focus on elucidating the impact of base pairing and the influence of the local environment on pairing tendencies. Subsequently, we pretrain our model by combining representations of RNA sequences with structural analysis results, followed by further training based on pairing constraints. To validate the robustness of our algorithm, extensive experiments were conducted using benchmark datasets resulting in achieving state-of-the-art prediction accuracy. Furthermore, experiments encompassing areas such as classification of RNA sequence representations, pseudoknot prediction, chemical mapping activities, and binding affinity of riboswitches were performed to verify its effectiveness in predicting RNA structure. These findings underscore both the potential and innovation inherent in our approach toward accurate RNA structure prediction while highlighting its significance within the field of RNA analysis.

Materials and methods

Overview

This study presents a deep learning-centric approach named DSRNAFold for predicting the secondary structure of RNA by integrating sequence and structural information using a unique architecture. This architecture is specifically designed to efficiently handle both local and long-range nucleotide interactions, avoiding the reliance on predetermined thermodynamic parameters and enabling the autonomous identification of complex structural motifs (Fig. 1).

First, the DSRNAFold framework transforms the position of each nucleotide in an RNA sequence into a detailed representation via a deep learning network, capturing the sequence's inherent features. This process includes constructing a latent representation using a fully connected layer and introducing a noise mask to reduce the risk of overfitting. Subsequently, through deep CNN layers, the contextual representation at each specific position is extracted, highlighting the use of multiple residual blocks. These blocks are configured to improve the ability of the model to understand the context of a sequence. To ensure robust model training, random sampling is adopted, enabling a sufficient amount of contextual information to be extracted from the dataset. During the model

training phase, specific loss functions are defined based on the analysis of similarities between sequences and fragments, allowing the model to better understand and predict secondary structures within RNA sequences (Fig. 1A). This step emphasizes the importance of extracting rich contextual information from sequence data, aiming to improve the accuracy of RNA structure prediction through in-depth analysis.

Moreover, the DSRNAFold framework uses an integrated approach to RNA structure analysis, focusing on the impact of base pairing and the influence of the local environment on pairing tendencies. This method draws on the concept of local weighted linear regression and initially considers canonical base pairing constraints by assigning weights based on the number of hydrogen bonds. Furthermore, differing from traditional approaches that may overlook the effects of local structural context, our method delves into how the local environment within RNA secondary structures affects pairing trends. It is speculated that pre-paired neighboring bases may promote further pairing by favorably altering the spatial configuration of RNA [35]. The process of calculating base pairing probabilities incorporates Gaussian probability distributions to model the decay of influence with spatial proximity, enhancing our understanding of how adjacent bases contribute to the stability and likelihood of base pairing (Fig. 1B). By employing specific weights, this approach emphasizes the role of spatial proximity in determining pairing likelihoods, aiming to overcome the issue of data sparsity while enriching the base pairing data, leading to a comprehensive and dimensionally rich structural representation and providing deep insights into RNA structures.

The results of RNA sequence representation and structural analysis are combined, and DSRNAFold pretrains on RNA sequences. First, the sequence representation module converts an RNA sequence into a representation vector, which is then expanded into a three-dimensional tensor for processing by a U-Net network architecture, generating a probability prediction matrix. A stringent constraint module is applied to this matrix to derive the RNA base pairing probability. To enhance the prediction accuracy and efficiency, DSRNAFold employs the dice and binary cross-entropy loss functions, leveraging their complementary strengths (Fig. 1C). The dice loss is adept at handling the probabilistic nature of binary classification, promoting model confidence and accuracy by directly targeting the discrepancy between the predicted probabilities and actual labels. The binary cross-entropy loss further improves the prediction accuracy and efficiency by providing a precise, sample-level assessment of the prediction error. By introducing a weighting coefficient, these loss functions are combined into a unified loss function, allowing flexible adjustment of the loss components' influence on the training process, thereby enhancing the model's ability to capture the nuances of RNA secondary structure prediction.

In the postprocessing stage, a constraint-based approach is used to improve the accuracy of RNA secondary structure prediction. By integrating the U-Net model with specific constraint-based methodologies, the importance of implementing particular base pairings and spatial constraints to enhance the biological accuracy is emphasized. Specifically, two primary constraints are enforced: a minimum separation requirement for base pairings and a limitation on the pairing capacity of each base. These constraints are applied to the predicted probability matrix to generate a more precise output that is closely aligned with canonical RNA structure rules.

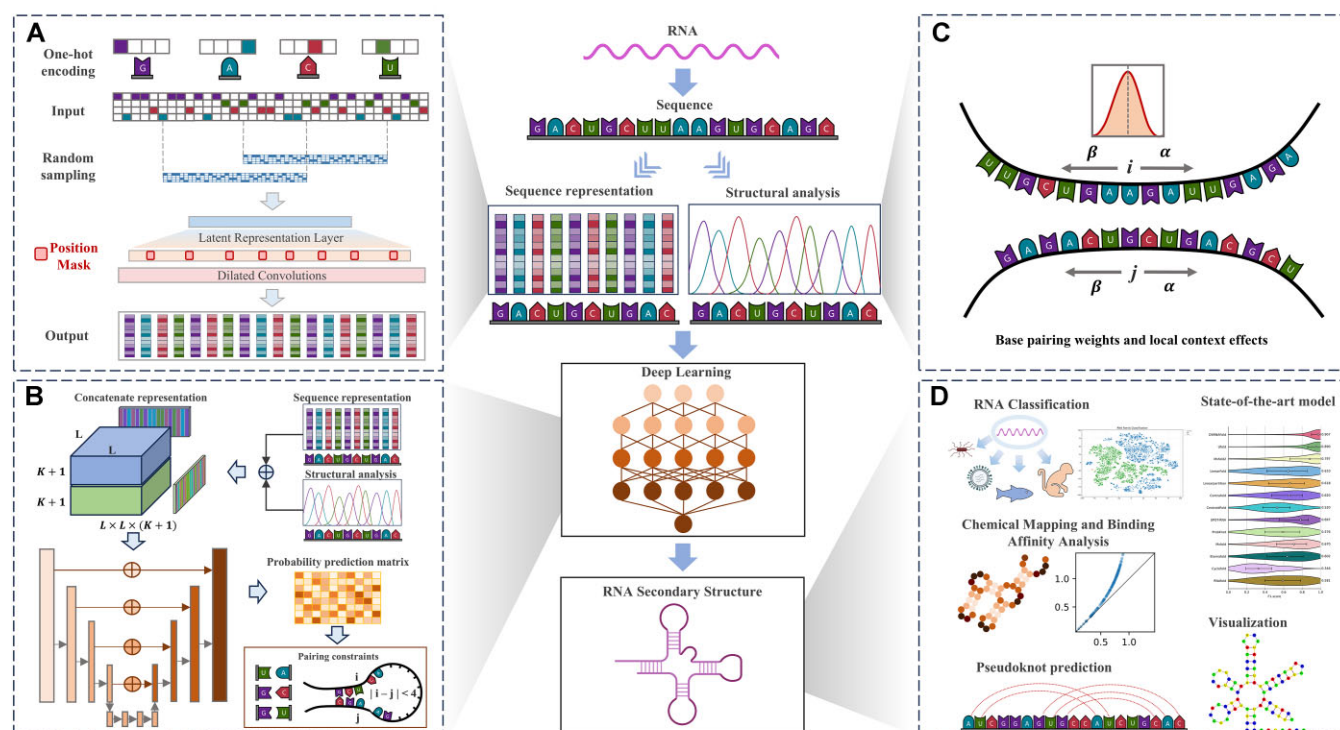


Figure 1. Overview of the DSRNAFold framework. The DSRNAFold framework provides a comprehensive overview for RNA secondary structure prediction. **(A)** Sequence representation: In the initial phase, each nucleotide's position is transformed into a detailed representation via a deep learning network, capturing inherent sequence features and reducing overfitting with a noise mask. **(B)** Structural analysis: RNA structure analysis is achieved by integrating concepts of locally weighted linear regression to enhance our understanding of spatial configurations and pairing likelihoods. **(C)** Prediction and postprocessing: This step involves model pretraining, generating prediction matrices through the U-Net architecture, and applying constraints for accurate RNA base pairings, utilizing loss functions for improved prediction accuracy. **(D)** Validation of results: This part displays the application of the DSRNAFold model to RNA sequence classification, pseudoknot prediction, chemical mapping, riboswitch binding affinity, and RNA secondary structure visualization to showcase its functionalities in a practical setting. It includes a comparative analysis with other existing methods, illustrating differences in approach and efficiency.

The $F1$ loss function is employed for model training and optimization, aiming to balance precision and recall to improve the prediction accuracy. The predictive outcomes are evaluated using precision and recall as the key metrics to assess the model's predictive capabilities, and the $F1$ score is used as a metric to balance precision and recall. This approach provides a robust framework for assessing the effectiveness of RNA secondary structure prediction models, ensuring a comprehensive and precise performance evaluation.

We systematically conduct a series of experiments to assess its performance across various aspects of RNA structure prediction, including RNA sequence and family classification, pseudoknot prediction, chemical mapping reactivity prediction, and binding affinity prediction of riboswitches, as further confirmed by visualization analysis (Fig. 1D). The model, through efficiently embedding RNA sequences into high-dimensional spaces and precisely processing nucleotide interactions using deep learning techniques, exhibits its predictive accuracy. The performance of this model in predicting RNA secondary structures with pseudoknots can confirm its ability to interpret misfolded states and accurately reflect RNA molecular structural features. These achievements not only highlight the application potential of DSRNAFold in RNA structure prediction but also emphasize the importance of continuous innovation in the field of RNA structure analysis.

Datasets

In this study, we use several benchmark datasets to rigorously evaluate the predictive capabilities of various RNA structure prediction models. Our dataset selection spans a broad spectrum of RNA families and sequence complexities, enabling a comprehensive assessment.

The primary datasets include the RNAStralign dataset [36], with 37 149 unique sequences from 8 RNA families, and the ArchiveII dataset [37], consisting of 3975 sequences across 10 RNA families. These datasets are pivotal for assessing the model performance across a broad spectrum of RNA structures. Additionally, the bpRNA-1m dataset [38], which contains 102 318 sequences from 2588 families, is utilized for its extensive coverage of RNA sequences, providing a significant resource for model training and evaluation. Chemical mapping data, sourced from the RNA Mapping Database [39] and processed with the RDAT Kit for the Eterna Cloud Lab experiments, complement these sequence-based datasets. These data, with sizes ranging from 605 to 3378 constructs before filtering and from 101 to 1088 constructs after filtering, include 12 711 individual constructs. The use of these data allows for the examination of RNA secondary structures through chemical mapping, offering an additional layer of validation for the predictive capabilities of the models.

To benchmark the models, two experimental studies were conducted. These experiments are similar to those involving

MXfold2 [28]. The first study involves training models on the RNAStralign dataset and evaluating them on ArchiveII dataset. The second study involves training the same models on the bpRNA-1m training set (TR0) and evaluating them on the bpRNA-1m test set (TS0), following data processing and evaluation methodologies used in previous studies. This approach includes the removal of redundant sequences by the CD-HIT-EST program [40] with a cutoff threshold of 80% and a random split of datasets into training and testing subsets, ensuring a rigorous assessment of the models. The redundancy removal process is critical for preventing model overfitting and is applied uniformly across datasets. Specifically, the process was applied to the bpRNA-1m dataset, resulting in two subsets, TR0 for training and TS0 for testing. The ArchiveII dataset, devoid of redundancy, is solely used for testing to assess the model's accuracy and generalizability. The specific number of sequences in the dataset is summarized in [Supplementary Tables S1–S3](#) and [Supplementary Fig. S4](#). To enhance the robustness against “unseen” RNA families, we constructed a dataset based on RNA families in the latest version of the Rfam database (Rfam 15). Within each RNA family in the training set, we employed the CD-HIT-EST to filter out sequences with an identity exceeding 80%. Additionally, sequences longer than 512 nucleotides were removed from the training and testing. Finally, the training set contains a total of 13 707 RNA sequences, and the test set contains 7686 RNA sequences. Incorporating chemical mapping data from 24 datasets from Eterna “Cloud Labs,” encompassing 38 846 constructs ranging in length from 80 to 130 nucleotides, provides an additional dimension for model evaluation. This approach enables a comprehensive analysis, ensuring that the models are well tested across diverse RNA sequences and that structural insights are gained from chemical mapping.

Deep learning-driven embedding for obtaining position-specific representations of RNA sequences

These representations of RNA sequences are designed to capture both short- and long-range interactions relevant to RNA folding. For example, certain consecutive nucleotides may form local motifs (e.g. hairpin loops), whereas distant bases might form canonical base pairs (A–U, G–C, G–U). By mapping each position to a high-dimensional embedding, the model can incorporate such biologically meaningful cues, bridging raw sequence information with the structural features observed in known RNA molecules. Here, RNA sequence data, conceptualized as sequential data, are delineated as $X = x_1, x_2, \dots, x_L$, where each x_i represents the i th position of a nucleotide and L denotes the sequence length. Each x_i is encoded into its representation $\gamma_i \in \mathbb{R}^{L \times K}$, where K is the dimension of the representation γ_i . The further details on the encoding process are shown in [Supplementary Fig. S1](#).

During the training phase for RNA sequence representation, RNA samples are generated using a random sampling strategy. This method ensures that any two segments of the original sequence data overlap, and the overlapping region serves as a sample for training the representation model. Specifically, when RNA sequence segments $S_m = \{x_m\}$ and $S_n = \{x_n\}$ are randomly sampled, their overlapping region $S_m \cap S_n$ is used for training the representation model. This approach, inspired by TS2Vec [41], allows the overlapping regions, which may represent distinct substructures, to contribute to the RNA sequence representation learning process.

In the model training phase, the primary objective is to substantially minimize the differences between the contextual representations on the overlapped segments, thereby ensuring the holistic representation of the RNA sequence data. Base comparison, which focuses on the differences in information present at each base position where segments of a single sample overlap, is defined as follows:

$$\text{BaseLoss}_{(m,u)} = -\log \frac{\exp(\gamma_{m,u} \cdot \gamma'_{m,u})}{\sum_{v \in P} \exp(\gamma_{m,u} \cdot \gamma_{m,v})}, \quad (1)$$

where P represents the base position set of segment m that overlaps. In this context, $\gamma_{m,u}$ is the representation of the overlapped region of segment m at base position u , and $\gamma'_{m,u}$ denotes the representation of the overlapping region of another segment that overlaps with segment m at base position u .

In parallel, sample comparison is defined to evaluate the difference in information within and between samples at the same position, enhancing the depth of the analysis.

$$\text{SamLoss}_{(m,u)} = -\log \frac{\exp(\gamma_{m,u} \cdot \gamma'_{m,u})}{\sum_{n \in S} \exp(\gamma_{m,u} \cdot \gamma_{n,u})}. \quad (2)$$

Here, S represents the sample set. The loss function for sample comparison is utilized to calculate the information loss between samples at the same position. The overarching loss function of the model is defined based on both base and sample comparisons, ensuring a holistic and comprehensive approach to the analysis of RNA sequence data.

$$\text{Loss} = -\frac{1}{|S| |P_S|} \sum_{m \in S} \sum_{u \in P_S} (\text{BaseLoss}_{(m,u)} + \text{SamLoss}_{(m,u)}). \quad (3)$$

In this framework, S represents the sample set and P_S denotes the position set of the sample, with $|S|$ and $|P_S|$ representing the number of elements in the respective collections. It ensures that biologically important motifs are represented consistently in the learned embedding.

Integrative RNA structure analysis with base pairing weights and local context effects

Inspired by the concept of local weighted linear regression and similar to CDPfold [42], we characterize RNA base sequences by considering the influence of base pairing. Initially, canonical pairing constraints are considered, and weights are assigned based on the number of hydrogen bonds:

$$P_{(x_i, x_j)} = \begin{cases} 3 & (x_i = G, x_j = C) \text{ or } (x_i = C, x_j = G) \\ 2 & (x_i = A, x_j = U) \text{ or } (x_i = U, x_j = A) \\ 1 & (x_i = G, x_j = U) \text{ or } (x_i = U, x_j = G) \end{cases}. \quad (4)$$

Beyond individual base pairs, we examine the impact of the local environment within RNA secondary structures on pairing tendencies. The propensity of a base to pair is theoretically influenced by the pairing status of adjacent bases [35], suggesting that pre-paired neighbors may promote further pairing by favorably altering the RNA's spatial configuration. This insight leads us to incorporate the local context into our analysis, acknowledging that RNA folding is not solely determined by isolated interactions but is significantly influenced by the surrounding nucleotide environment. To achieve quantification, we refine the calculation of pairing probabilities between positions i and j , incorporating the influence of adjacent pairings. This refinement is captured by the following formula:

$$M_{(x_i, x_j)} = P_{\text{left}} + P_{(x_i, x_j)} + P_{\text{right}}, \quad (5)$$

where P_{left} and P_{right} account for the pairing probabilities adjacent to i and j , respectively. Gaussian functions serve as weights, accentuating the role of proximity in determining pairing likelihoods. This approach mitigates the issue of sparse data and augments the richness of base pairing data, culminating in a comprehensive representation dimensionality of $L \times L \times (K + 1)$ (see [Supplementary Fig. S3](#)).

Optimizing RNA secondary structure prediction

In deep learning-driven embedding, an RNA sequence of length L undergoes two simultaneous transformations. First, it is processed by the sequence representation module to generate a representation vector with dimensions $L \times K$, capturing the intrinsic properties of the sequence. Second, structural analysis is performed on the sequence, resulting in a distinct representation vector with dimensions $L \times 1$ that encapsulates its structural characteristics. These vectors are then concatenated to form a three-dimensional tensor with dimensions $L \times L \times (K + 1)$, effectively leveraging both sequence and structural information for comprehensive analysis. The tensor is processed through a U-Net network architecture, resulting in a probability prediction matrix of dimensions $L \times L$. The final RNA secondary structure prediction is derived by applying a stringent constraint module to this matrix (see [Supplementary Fig. S2](#)).

The model incorporates both the dice loss and binary cross-entropy loss functions in a unified framework. The integration of these loss functions for RNA secondary structure prediction exploits their complementary benefits. The dice loss is adept at handling the probabilistic nature of binary classification, promoting model confidence and accuracy by directly targeting the discrepancy between the predicted probabilities and actual labels. This is particularly useful in the context of RNA secondary structure prediction, where the model must accurately distinguish between structured and unstructured regions within the RNA sequence. By incorporating the dice loss into the model's training process, we can ensure that predictions are correct not only in terms of overall structure but also in terms of the specific arrangement of base pairs. The smooth parameter is introduced in the dice loss to prevent division by zero during computation. Dice loss is given by

$$L_{\text{Dice}} = 1 - \frac{2 \cdot \sum_{i=1}^N (y_i \cdot \text{sigmoid}(x_i)) + \text{smooth}}{\sum_{i=1}^N \text{sigmoid}(x_i) + \sum_{i=1}^N y_i + \text{smooth}}, \quad (6)$$

where x represents the output of the model, y represents the label, and smooth is set 1. The binary cross-entropy loss further improves the model confidence and accuracy by offering a precise, sample-wise assessment of the prediction error, focusing on the inherent nuances of the binary classification task. This loss function is defined as follows

$$L_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\text{sigmoid}(x_i)) + (1 - y_i) \cdot \log(1 - \text{sigmoid}(x_i))]. \quad (7)$$

To integrate the loss functions into a cohesive framework for RNA secondary structure prediction, we propose a unified loss function that combines these two components with a weighting coefficient. The combined loss function is defined as

$$\text{Loss} = \lambda_1 L_{\text{Dice}} + \lambda_2 L_{\text{BCE}}, \quad (8)$$

where λ_1 and λ_2 are weight coefficients that balance the contribution of each loss component and are both set to 0.5 in this study. This formulation allows for flexible adjustment of the influence of the loss components on the training process, potentially enhancing the model's ability to capture the nuances of RNA secondary structure prediction.

Constraint-based approaches for RNA secondary structure prediction

We integrate a U-Net model with constraint-based methodologies, focusing on enhancing the biological accuracy through specific base pairing and spatial constraints. We enforce two primary constraints: a minimum separation requirement for base pairing and a limitation on the pairing capacity of each base, inspired by the approach used in E2Efold [25]. These constraints are meticulously applied to the predicted probability matrix, yielding a refined output that closely aligns with canonical RNA structure rules. For model training and optimization, the $F1$ loss function is employed, emphasizing the balance between precision and recall to improve the prediction accuracy. This function is defined as:

$$F1_{\text{loss}} = -\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (9)$$

The aim is to minimize prediction errors while maximizing the model's ability to accurately identify relevant pairings. Further elaboration on these methodologies and their implementation is available from [Article IV in Supplementary Section S1](#), which provides a comprehensive overview of our approach to RNA structure prediction.

Performance measures

To assess each experiment, the predictive outcomes for base pairs were categorized into four distinct scenarios: true positive (TP), false positive (FP), true negative, and false negative (FN). The classification results were systematically compiled into a confusion matrix. To evaluate the model's accuracy, we incorporated the precision ratio (Precision) and recall ratio (Recall) as key metrics to quantify the model's predictive capabilities. These metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (10)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (11)$$

where TP denotes the number of base pairs accurately predicted as true positives, FP represents the incorrectly predicted base pairs as false positives, and FN indicates the number of base pairs in the reference structure that were missed in the prediction, categorized as false negatives. Additionally, the $F1$ score was employed as a balanced metric between precision and recall, calculated as their harmonic mean, as defined in formula (12).

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (12)$$

The $F1$ score highlights the model's overall efficacy with higher values indicating superior performance. This comprehensive approach, which integrates precision, recall, and F values, provides a robust framework for evaluating the model's effectiveness in RNA secondary structure prediction, ensuring a thorough and accurate performance assessment.

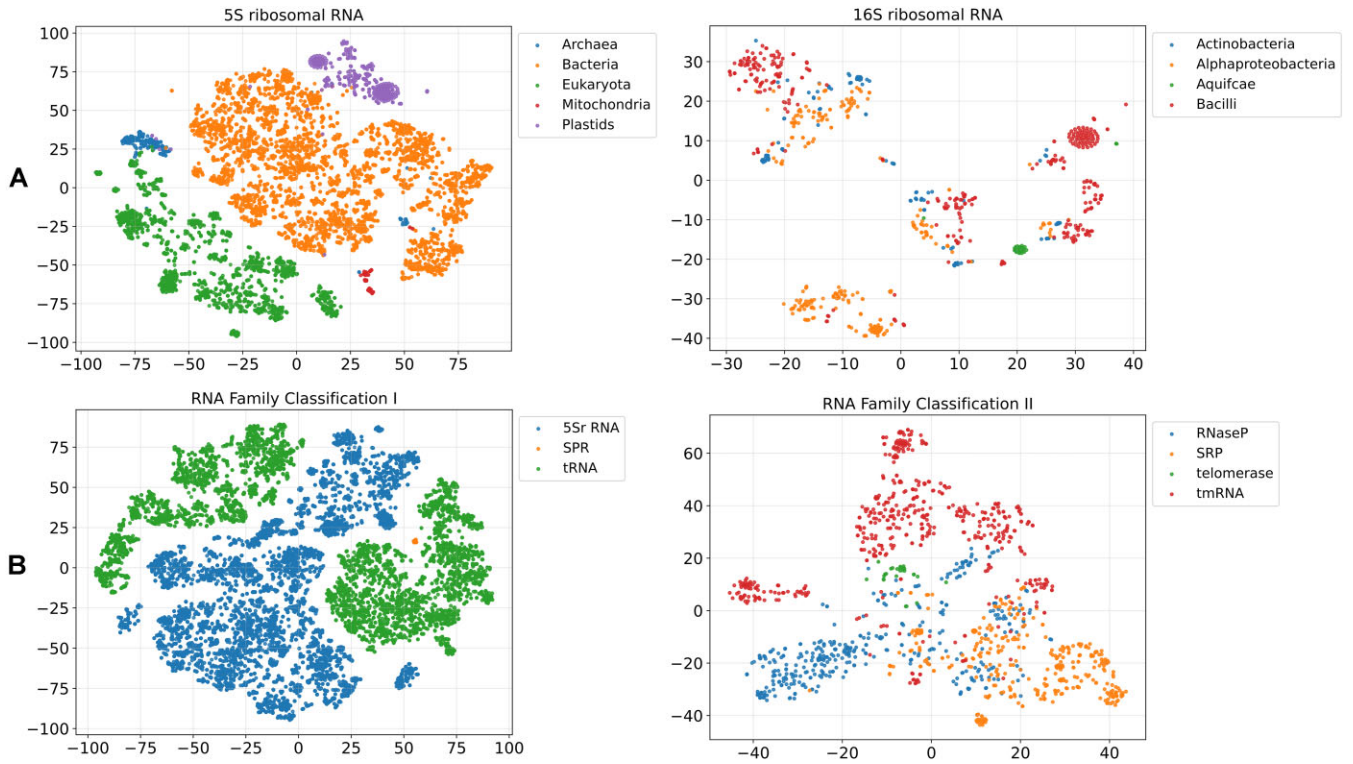


Figure 2. Visualizations of RNA sequence representations with phylogenetic and familial classification patterns. **(A)** t-SNE visualization of 5S rRNA and 16S rRNA, colored with phylogenetic information. Each point is colored based on its phylogenetic origin, illustrating distinct clusters that correspond to different biological taxa such as Archaea, Bacteria, Eukaryota, Mitochondria, and Plastids. The data for these visualizations were obtained from RNAStralign and analyzed using t-SNE to highlight phylogenetic relationships. **(B)** t-SNE distribution of RNA family classifications. RNA sequences are grouped by family, including 5S rRNA, SRP RNA, tRNA, RNaseP, telomerase, and tmRNA. Points are colored to represent different RNA families, demonstrating clear spatial separations that reveal the model’s capability to discriminate among various RNA types based on structural attributes. The sequence data were similarly sourced and analyzed using t-SNE to explore and validate the model’s effectiveness in sequence classification

Results

Discriminative capacity of RNA sequence representation

To investigate the distinctive features and classifications of various RNA types, which is essential for elucidating the subtle features unique to different RNA types and gaining a deeper understanding of RNA sequence characteristics, the efficacy of the proposed model’s sequence characterization was rigorously assessed through t-distributed stochastic neighbor embedding (t-SNE) visualizations. They successfully revealed distinct clusters that are aligned with phylogenetic and familial classifications, demonstrating the model’s precise capability to differentiate between the RNA types based on their sequence characteristics.

Clusters are observed in accordance with biological taxa, including Archaea, Bacteria, Eukaryota, Mitochondria, and Plastids, for 5S ribosomal RNA (rRNA), and more granular classifications are observed among bacterial 16S rRNA (Fig. 2A). The pronounced grouping of these taxa underscores the model’s ability to capture phylogenetic signatures within its representational space, confirming the evolutionary relevance of the sequence representation. Furthermore, the categorization by RNA families reveals marked segregation among 5S rRNA, Signal Recognition Particle RNA (SRP RNA), transfer RNA (tRNA), RNaseP, telomerase, and transfer-messenger RNA (tmRNA) (Fig. 2B). The model’s ability to encode family-specific structural attributes is evidenced by the clear spatial separation among the RNA families. This distinc-

tion is vital for understanding the model’s discriminative power.

In addition, we quantitatively examined whether the latent vectors also encode fine-grained structural information. For two representative Rfam families (RF00006 and RF00254; sequence length ≤ 128 nt), we computed pair-wise negative Euclidean distances in the embedding space and compared them with secondary structure similarity scores obtained with RNAforester [43], a structure-aware alignment tool. Spearman correlation analysis revealed positive and significant associations between embedding similarity and structural similarity ($\rho = 0.569$ for RF00006 and $\rho = 0.860$ for RF00254). These results indicate that the learned embeddings capture not only sequence-level signals but also structural features that are consistent with experimentally derived secondary structure relationships, thereby reinforcing the biological relevance and discriminative capacity of the representation.

The defined clusters across all visualizations indicate that the model’s learned representation is both robust and biologically pertinent, encapsulating essential sequence motifs and structural configurations that are reflective of both phylogenetic relationships and cellular functions. These visualizations, therefore, validate the model’s potential as an effective tool for RNA sequence identification and classification.

Comparison with existing methods

To comprehensively evaluate the performance of DSRNAFold, we benchmarked it against a wide range of estab-

lished RNA secondary structure prediction methods, including UFold [27], Sincfold [44], MXfold2 [28], LinearFold [29], LinearPartition [45], CONTRAfold [31], CentroidFold [30], SPOT-RNA [24], ProbKnot [46], EternaFold [47], CycleFold [48], and RNAfold [18]. While SPOT-RNA2 [49] represents an important advance by integrating evolutionary features, it was excluded from direct comparison as all models analyzed here are based solely on sequence-derived features. For the comprehensive assessment, two primary datasets were utilized. First, RNAStralign was used for training, and ArchiveII was used for testing. Then, TR0 and TS0 were used for training and testing, respectively. Our analysis focused on the recall, precision, and *F1* score to provide a holistic view of each model's performance in terms of accuracy and robustness.

As shown in Table 1, DSRNAFold achieved competitive or superior performance compared with existing methods across both benchmarks. On the RNAStralign–ArchiveII setting, DSRNAFold demonstrated an *F1* score of 0.907, comparable to Sincfold (0.923), and markedly higher than traditional thermodynamics-based models such as RNAfold and LinearPartition. Models such as LinearFold and CONTRAfold performed moderately, while CycleFold exhibited the lowest performance (0.346). Further family-specific analysis based on the heatmap in Fig. 3A revealed that DSRNAFold consistently achieved top or near-top *F1* scores across most RNA families. DSRNAFold demonstrated outstanding performance on 5S (0.978) and tmRNA (0.953), and remained competitive in challenging families like RNP (0.870) and telomerase (0.869). Sincfold showed slightly better results than DSRNAFold for RNP and tmRNA families but underperformed in the 16S rRNA and grp1 families. UFold exhibited robust performance on 23S rRNA and grp1, although with a slight drop in telomerase structures. Traditional thermodynamics-based models, such as LinearFold and RNAfold, generally achieved lower *F1* scores across families, particularly struggling on complex long-range structures like 5S and tmRNA. CycleFold consistently showed the lowest *F1* values across all families. On the bpRNA-1m (TR0–TS0) dataset (Table 1), the overall *F1* scores decreased due to the dataset's higher complexity. DSRNAFold still achieved the best performance (*F1* = 0.627), narrowly outperforming Sincfold (0.624) and UFold (0.608). Family-level analysis on TS0 further revealed that DSRNAFold excelled in the SPR (0.927) and RNP (0.663) families, maintaining high prediction robustness for structured RNAs. Sincfold outperformed DSRNAFold slightly on RNP (0.742) but lagged behind on SPR. UFold showed competitive results on SPR (0.948) but decreased performance on RFAM (0.430). The comparative results demonstrate that DSRNAFold maintains strong and stable predictive performance across different datasets and RNA families, highlighting its generalization ability.

We evaluated the performance of DSRNAFold on previously unseen RNA families. It is widely observed that learning-based methods tend to perform poorly on these RNAs, as they are not represented in the training data [27]. To address this challenge, hybrid methods that integrate free energy minimization with deep learning approaches, such as MXfold2, have been developed. However, these methods inadvertently introduce biases into the prediction models, which may reduce their performance on within-family RNAs. Given that the majority of RNA secondary structure prediction methods still rely on energy-based models, we selected several representative approaches for comparison, including UFold, Sinc-

Table 1. Comparison of DSRNAFold with existing methods

Method	RNAStralign–ArchiveII			bpRNA-1m (TR0–TS0)		
	Recall	Precision	<i>F1</i>	Recall	Precision	<i>F1</i>
DSRNAFold	0.896	0.927	0.907	0.649	0.641	0.627
Sincfold	0.953	0.908	0.923	0.702	0.603	0.624
UFold	0.895	0.893	0.890	0.689	0.564	0.608
MXfold2	0.793	0.806	0.767	0.598	0.500	0.529
LinearFold	0.607	0.629	0.610	0.581	0.517	0.530
Linearpartition	0.623	0.623	0.618	0.629	0.509	0.547
CONTRAfold	0.652	0.597	0.620	0.655	0.482	0.541
CentroidFold	0.437	0.689	0.520	0.516	0.549	0.508
SPOT-RNA	0.731	0.694	0.697	0.693	0.560	0.603
ProbKnot	0.603	0.557	0.576	0.634	0.442	0.507
MXfold	0.673	0.673	0.670	0.639	0.500	0.546
EternaFold	0.636	0.576	0.602	0.663	0.475	0.539
Cyclefold	0.442	0.286	0.346	0.498	0.240	0.314
RNAFold	0.614	0.555	0.581	0.631	0.446	0.508

Note: Numbers in bold represent the highest scores in each performance category for both datasets, highlighting the model that achieved the best results in terms of recall, precision, and *F1* score.

fold, MXfold2, and LinearPartition. For evaluation, RNA sequences were divided into two groups based on their lengths: 128 and 256 bp. The results (see [Supplementary Table S10](#)) demonstrate that DSRNAFold achieves performance comparable to other methods like MXfold2, all of which incorporate thermodynamic constraints in their objective functions. DSRNAFold is purely a learning-based method; it does not utilize data augmentation techniques, as seen in UFold, nor does it incorporate free energy minimization strategies, yet it was able to predict structures of RNAs not represented in its training data.

In addition, to verify the effectiveness of the proposed components in DSRNAFold, we present a comparison between the DSRNAFold model and its variants on the ArchiveII and TS0 datasets (see [Supplementary Tables S4–S6](#)), respectively. It verifies the effectiveness of the structural analysis and the sequence representation we proposed. The results indicate that all the aforementioned components of DSRNAFold are indispensable. The computational efficiency and resource requirement analysis of these models is summarized in [Supplementary Tables S8 and S9](#).

Prediction of secondary structures with pseudoknots

Pseudoknots represent one of the most complex classes of RNA secondary structures, formed through base pairing between nucleotides in a hairpin loop and an external region, resulting in intercalated helical arrangements. Accurate prediction of pseudoknots remains a significant challenge due to their computational complexity and the NP-complete nature of energy minimization frameworks for such structures. Unlike certain heuristic-based methods that explicitly model pseudoknots, DSRNAFold adopts a unified contact probability learning strategy, treating pseudoknots and nonpseudoknots without distinction during training.

To specifically evaluate DSRNAFold's capability in handling pseudoknots, we constructed a pseudoknot-focused benchmark dataset derived from the bpRNA database. Sequences containing at least one pseudoknot element were selected, redundancy was reduced using CD-HIT-EST with an 80% identity threshold, and the dataset was divided into

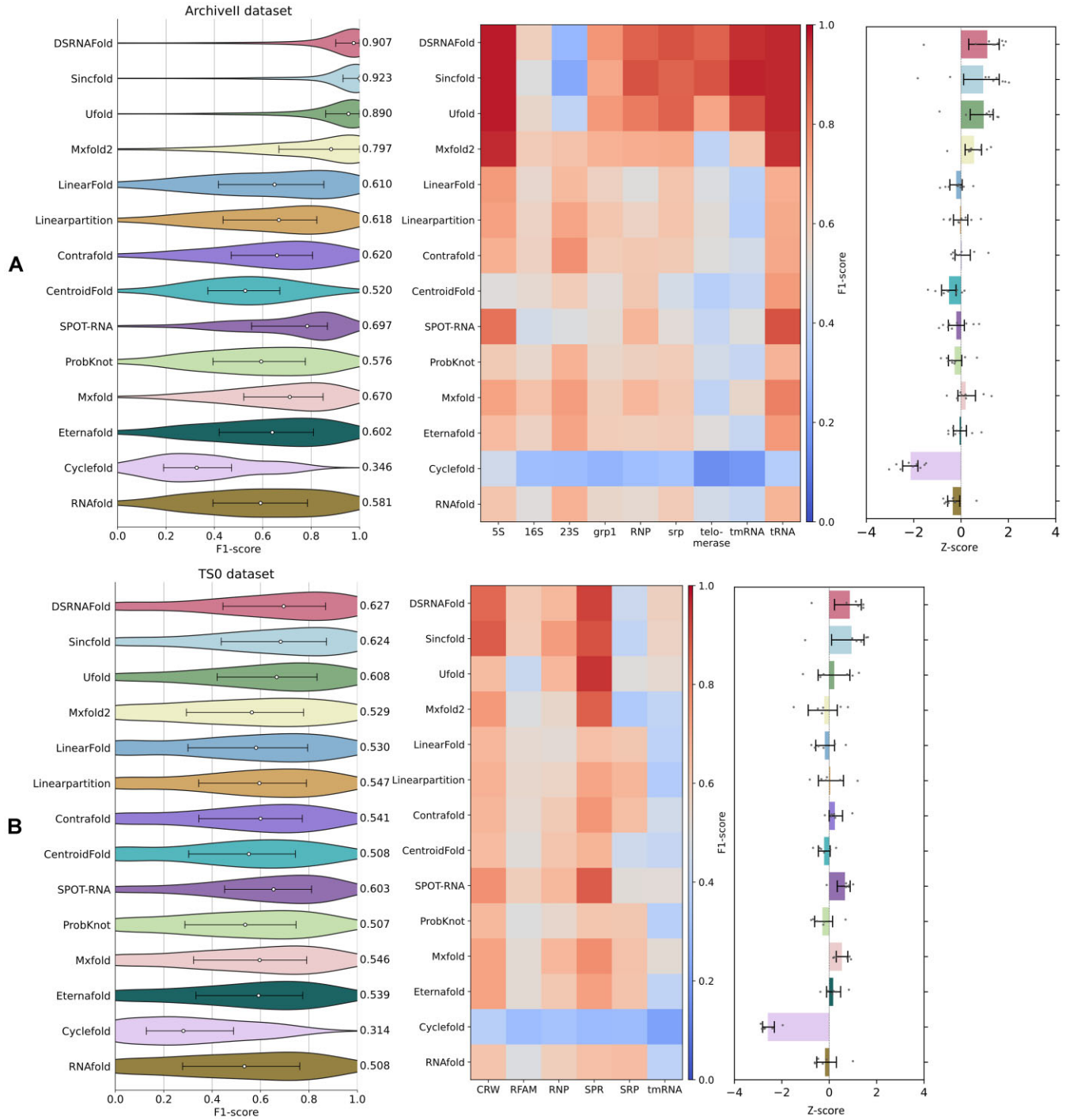


Figure 3. Performance comparison with existing methods. **(A)** Performance on RNAstralign and Archivel. It provides a detailed comparison of average $F1$ scores for DSRNAFold against established models. The left side features violin plots that summarize the distribution of $F1$ scores across all models. The center and right panels detail performance across nine RNA families, showcasing the range of $F1$ scores and z-scores, respectively, to assess statistical significance in performance differences among the models. Data for this analysis were obtained from testing and training on the RNAstralign and Archivel datasets. **(B)** Performance on bpRNA-1m (TR0-TS0). It uses the bpRNA-1m (TS0) dataset to present similar performance metrics. The left side again features violin plots for $F1$ scores, while the center and right panels provide a breakdown of $F1$ scores and z-scores by family, respectively. This consistent approach allows for a direct comparison of model performance under different experimental conditions and across a broader spectrum of RNA families, emphasizing the robustness and adaptability of DSRNAFold.

Table 2. Prediction results for a secondary structure with pseudoknots

Model	All base pairs			Pseudoknot base pairs		
	Precision	Recall	F1 score	Precision	Recall	F1 score
DSRNAFold	0.635	0.562	0.588	0.267	0.318	0.279
Sincfold	0.515	0.739	0.587	0.284	0.241	0.25
UFold	0.529	0.574	0.547	0.2	0.286	0.229
SPOT-RNA	0.581	0.524	0.541	0.177	0.168	0.168
ProbKnot	0.381	0.419	0.393	0.014	0.009	0.011
IPknot	0.449	0.409	0.419	0.026	0.024	0.025

Note: Numbers in bold represent the highest scores in each performance category for both datasets, highlighting the model that achieved the best results in terms of recall, precision, and F1 score.

TRpdn (training) and TSpdn (testing) sets at an 8:2 ratio. We compared DSRNAFold against several established deep learning and thermodynamics-based methods, including Sincfold, Ufold, SPOT-RNA, ProbKnot, and IPknot. As summarized in Table 2, DSRNAFold achieved the highest F1 score on TSpdn, indicating strong generalization even under complex structural constraints. Among the compared methods, Sincfold achieved a relatively high recall but lower precision, suggesting a tendency to overpredict base pairs. Ufold and SPOT-RNA exhibited moderate precision but lower recall, reflecting challenges in capturing all relevant base pair interactions. ProbKnot and IPknot, as thermodynamics-based methods explicitly modeling pseudoknots, showed considerably lower overall performance, highlighting the difficulty of heuristic approaches in complex datasets.

To isolate model performance specifically on pseudoknotted base pairs, we further extracted and evaluated only the pseudoknot regions. As shown in Table 2, DSRNAFold again achieved the highest F1 score. Sincfold displayed a slightly higher precision on pseudoknots compared to DSRNAFold but suffered from lower recall, indicating that it missed a substantial proportion of pseudoknotted interactions. Ufold and SPOT-RNA demonstrated limited predictive capacity for pseudoknots, while ProbKnot and IPknot performed poorly, consistent with the known limitations of heuristic methods in handling complex pseudoknotted topologies.

The DSRNAFold model stands out as an exceptionally promising tool in the landscape of RNA secondary structure prediction, with particular efficacy in accurately predicting structures that encompass pseudoknots. The performance of this model is superior in terms of the precision and F1 score, surpassing those of the other models (Fig. 4). These results suggest that although DSRNAFold does not explicitly model pseudoknots through specialized architectural components, its learning framework implicitly captures relevant structural patterns, leading to superior predictive performance. In contrast, heuristic pseudoknot modeling strategies, while efficient in specific contexts, often fail to generalize to diverse and complex pseudoknotted configurations.

Predicting chemical mapping reactivities

We assessed the efficacy of our RNA secondary structure prediction model to interpret data from chemical mapping experiments, which is widely acknowledged for its pivotal role in elucidating RNA secondary structures [50–52]. Chemical mapping, which is utilized extensively on the Eterna online platform, has become a fundamental method for structural

determination, offering comprehensive insights into the structural nuances of RNA [53]. This technique reveals the reactivity patterns of nucleotides to chemical modifications, providing an averaged view of their accessibility and suggesting their protection status through base pairing or interactions with binding partners [54].

To confirm the practicality and replicability of our approach, particularly in capturing the array of misfolded states disclosed by chemical mapping, we used data from Eterna “Cloud Labs.” This distinctive dataset includes 38 846 constructs designed by participants, ranging from 80 to 130 nucleotides in length, produced through iterative cycles on the Eterna platform. The chemical mapping data were refined by removing sequences with over 80% redundancy and excluding shared 3′ primer binding sites. After refinement [47], the dataset included 12 711 constructs, varying from 101 to 1088 in size, demonstrating consistent distributions in GC content, sequence length, and loop numbers within target structures, thus ensuring data integrity. These constructs underwent periodic *in vitro* mapping using selective 2′-hydroxyl acylation analyzed by primer extension and multiplexed accessibility probing read out through sequencing chemical mapping protocols [39].

The critical relationship between the probability of a nucleotide being unpaired and its measured reactivity in chemical mapping experiments forms the foundation for enhancing the accuracy of RNA structure predictions [47]. This relationship has historically been defined through various functional forms [55, 56]. In this study, we focused on quantitatively analyzing how well our secondary structure prediction model could replicate and interpret these chemical mappings, especially for identifying and interpreting extensive potential misfolded states. To achieve this, we employed the Pearson correlation coefficient between unpaired probability and experimentally measured reactivity as a measure of model quality.

In terms of the mean Pearson correlation coefficients across 24 datasets (Fig. 5A), we can observe a range of correlation values between the predicted RNA secondary structures and the chemical mapping data. These coefficients, which reflect the congruence between our model’s predictions and the experimentally determined structures, vary significantly across different rounds, defined as distinct batches of chemical mapping experiments. Higher correlation coefficients, as observed in rounds 1, 6, 7, 20, and 22, suggest a strong alignment between the model’s predictions and the chemical mapping results. This finding indicates that for these specific rounds, the model was able to capture the nuances of the RNA structure effectively, yielding predictions that mirror the reactivity patterns observed in chemical mapping, which are indicative of nucleotide accessibility and the RNA’s folding state. Conversely, the lower coefficients in rounds 18 and 19 indicate a weaker agreement between the predicted structures and the experimental data, and the correlation coefficients are mainly <0.2 (Fig. 5B). Descriptive statistics for rounds 00–23 are provided in Supplementary Table S7, showing no obvious outliers for rounds 18 and 19. This finding could be due to several factors, such as the inherent complexity of RNA folds in these rounds, which might not be fully captured by the prediction model, or potential disparities in the experimental conditions or data quality. Notably, the range of correlation coefficients spans from ~0.116 to 0.649, illustrating the variability in the model’s predictive performance (Fig. 6A and Supplementary Fig. S7). The mean correlation coefficient

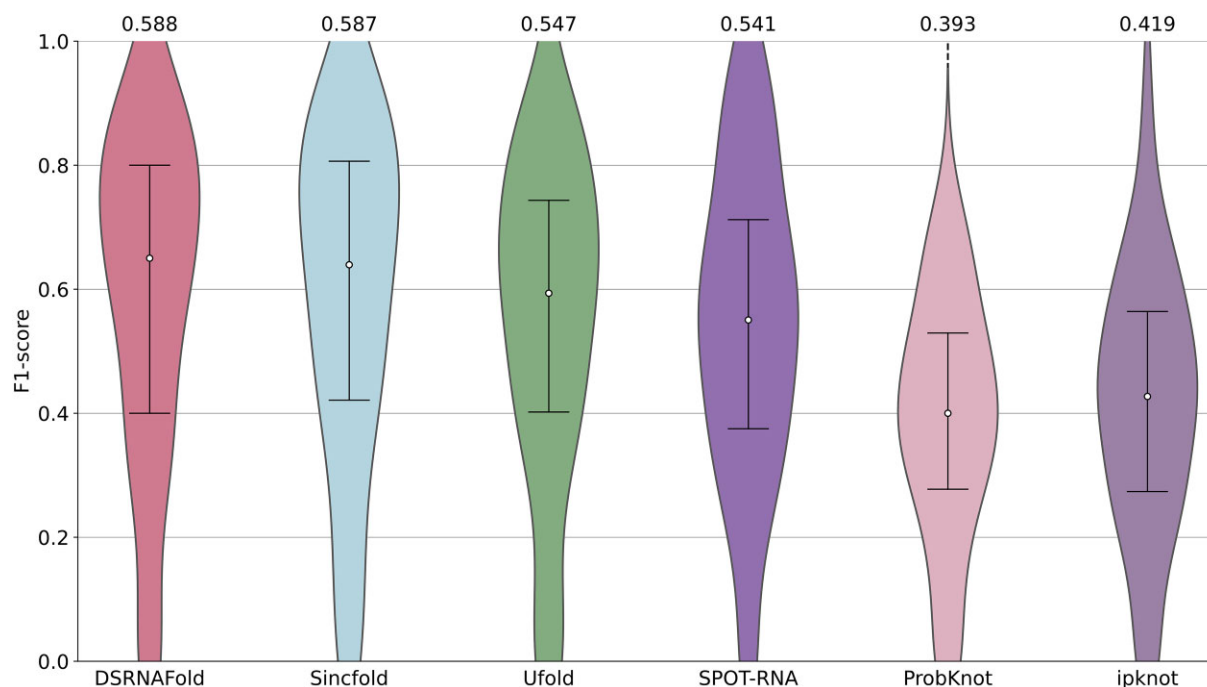


Figure 4. Evaluation results for RNA structures with pseudoknots. This figure presents violin plots comparing the *F1* scores of different models in predicting RNA secondary structures that include pseudoknots. Each plot displays the distribution of *F1* scores for the models tested. The numbers at the top of each plot represent the mean *F1* scores for each model, providing a quick reference for overall model performance in this specific task.

across all samples is 0.443, with a standard deviation of 0.219. The quarterly distribution of correlation values indicates that 25% of the rounds have a correlation of at least 0.609, which aligns with the upper quartile of the data and reflects the predictive success of the model. The median value of 0.456 implies that on half of the dataset, moderate-to-strong predictive alignment is achieved. High correlation values, particularly those within the upper quartile, underscore the model's ability to capture the intricate patterns of RNA folding and the consequent chemical reactivities. Compared with the reported test results [47], the performance of the proposed method is better than that of most of the nonenergy calculation methods (see [Supplementary Fig. S5](#)).

In this study, we conducted 24 experiments to explore the relationships between correlation coefficients and *z*-scores in detail. A significant positive correlation was observed between the correlation coefficients and *z*-scores, indicating that an increase in *z*-scores is associated with enhanced model performance. In most rounds, the data points were closely aligned along an upward trend, revealing a high degree of consistency and stability in model performance (Fig. 6B). However, deviations from the main trend line were observed in certain rounds, notably in rounds 2, 8, 19, and 21, which may suggest anomalous model behavior under specific conditions or in the presence of data abnormalities. Although these outliers might not be statistically significant, their presence alerts us to the necessity of further analyzing the robustness of the model under varying data distributions (Fig. 6C). Additionally, while a positive correlation was observed for all experiments, the magnitude of the correlation coefficient varied across rounds. This variation could reflect the impact of slight changes in the experimental conditions or heterogeneity in the input data on the model performance. These findings underscore the importance of conducting in-depth analyses to identify and understand fluctuations in model performance. Such insights are

crucial for improving the model and enhancing its adaptability and predictive accuracy in a broader range of application scenarios.

Predicting binding affinity of riboswitches

Riboswitches are RNA molecules that alter their structure upon binding to an input ligand, subsequently regulating various cellular processes such as transcription, translation, splicing, or the binding of a reporter molecule [57, 58]. According to the experimental strategy outlined in the previously published literature [47], we linked the probability that a riboswitch adopts a structural feature capable of binding its output to the experimentally measured binding affinity through relative ratios to those of a reference state. We evaluated the capability of DSRNAFold to predict the relative binding affinity of synthetic riboswitches to their output reporter, the fluorescently tagged MS2 viral coat protein, in the absence of an input ligand.

Similar to the chemical mapping datasets, each riboswitch dataset was filtered to exclude highly similar sequences. These riboswitches originated from two sources: the first set was designed by citizen scientists on Eterna [59], including 93-FMN, 95-FMN, 96-FMN, 97-FMN, 98-FMN, 107-FMN, 107-Trp, and 107-Theo. The second set consisted of riboswitches designed computationally using the RiboLogic package [60], probed concomitantly with Eterna riboswitches, including Ribologic-FMN, Ribologic-Trp, and Ribologic-Theo.

Unlike thermodynamic models, the DSRNAFold model, based on deep learning, cannot predict the energy changes caused by the binding of riboswitches to small molecule ligands. Consequently, we focused on predicting the relative binding affinity of fluorescently tagged MS2 viral coat proteins to riboswitch molecules in the absence of small molecule ligands. Given that the MS2 fluorescent protein binds to a

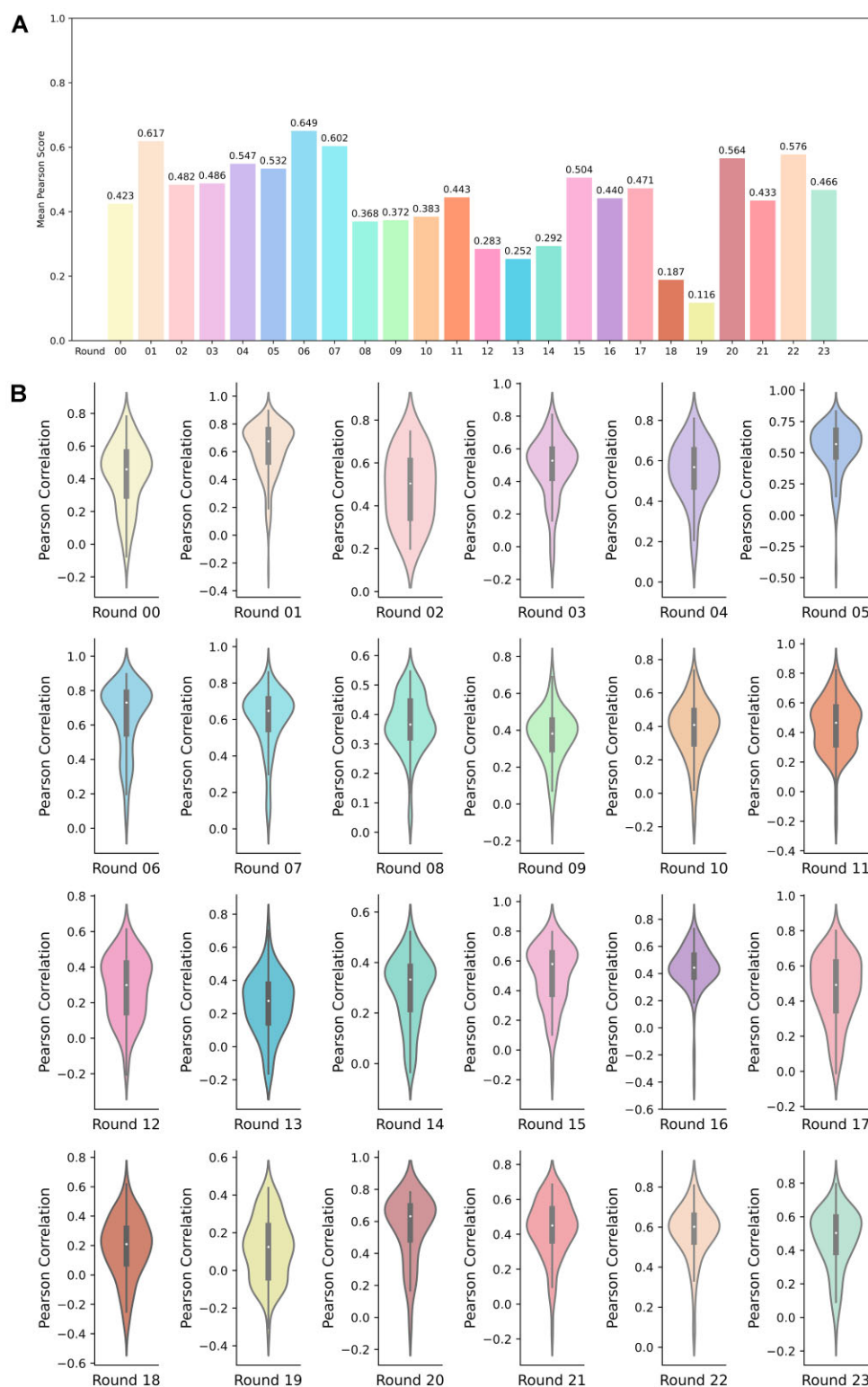


Figure 5. Evaluation results of chemical mapping reactivities. **(A)** Mean Pearson correlation coefficients across 24 datasets. It displays the mean Pearson correlation coefficients that quantify the alignment between predicted RNA secondary structures and experimentally measured chemical mapping reactivities. These coefficients are calculated from 24 distinct batches of chemical mapping experiments, referred to as “rounds,” conducted using data from the Eterna “Cloud Labs.” **(B)** Distribution of Pearson correlation coefficients. It shows the distribution of correlation coefficients across all rounds, highlighting variations and trends in the data. This visualization helps to depict the consistency and variability of the model’s performance in replicating the reactivity patterns observed in chemical mapping, with higher coefficients indicating better alignment between model predictions and actual chemical reactivities.

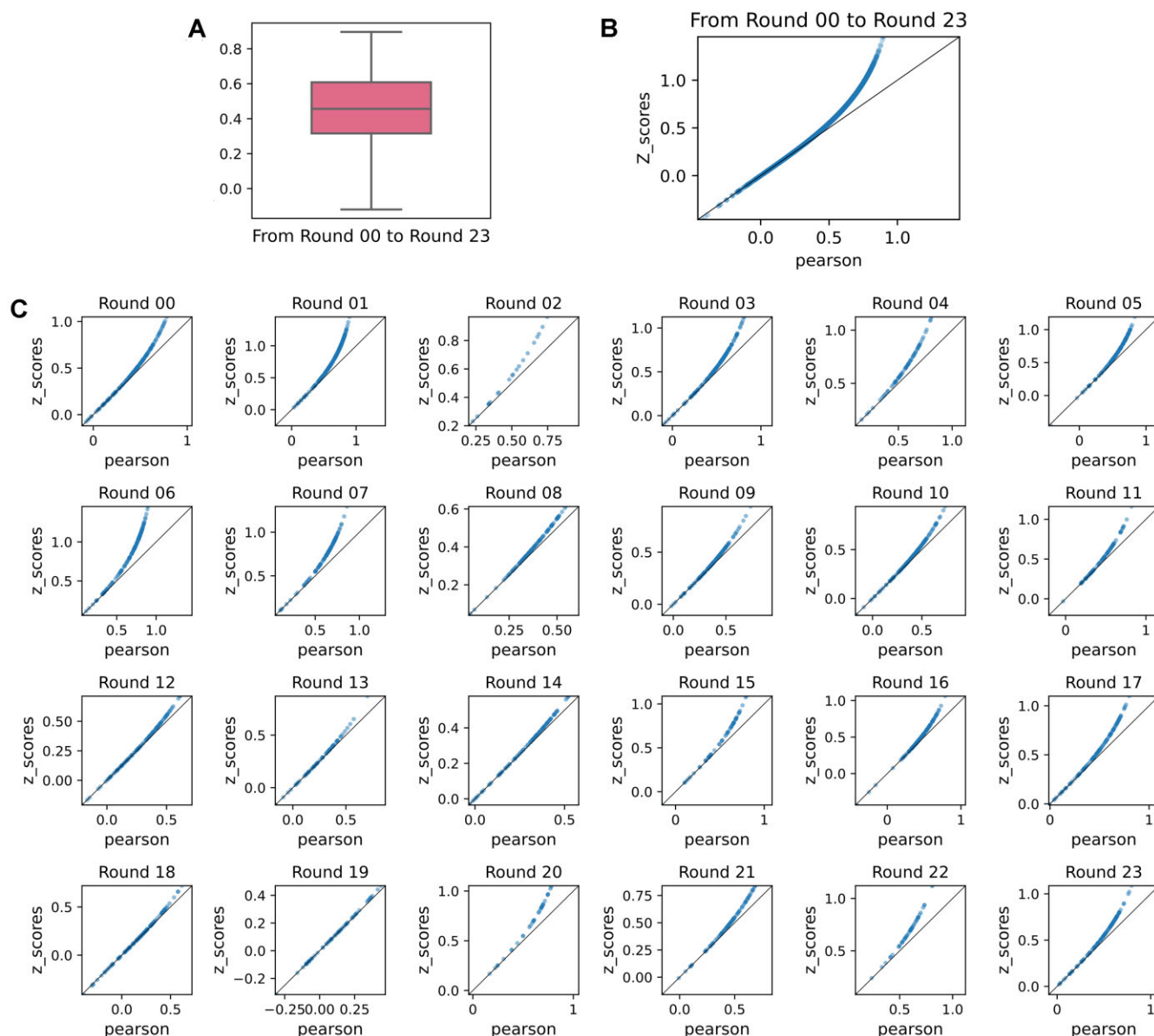


Figure 6. Correlation between RNA secondary structures and chemical mapping across experimental rounds. **(A)** Boxplot distribution of the Pearson correlation coefficients. This part shows a boxplot distribution of Pearson correlation coefficients across all 24 experimental rounds. The boxplots illustrate the range and variability of correlation values, indicating the degree of alignment between the model's predicted RNA structures and actual chemical mapping data. These coefficients were calculated to evaluate the model's accuracy in reflecting the nuanced chemical reactivities of RNA structures across different experimental setups. **(B)** Overall relationship between z-scores and Pearson correlation coefficients across all 24 datasets. This scatterplot displays the overall relationship between z-scores and Pearson correlation coefficients, offering insights into how well the model performs across a diverse range of chemical mapping data. **(C)** Round-by-round scatterplot analysis of z-scores and Pearson correlation coefficients. It provides a detailed round-by-round comparison, highlighting specific rounds where the model showed either strong or weak performance. It illustrates the consistency and stability of the model's predictive accuracy across various conditions, as well as identifying rounds with anomalous behavior.

specific hairpin loop structure, we only needed to predict the probability of formation of this constrained structure. Specifically, a riboswitch molecule is selected as a reference structure, and the ratio of the pairing probability of its terminal bases to the pairing probability of the terminal bases of other riboswitch constrained structures is used as a measure of binding affinity. Detailed theoretical derivation can be found in [47].

The experiment involved 11 datasets, totaling 7107 sequences. To better evaluate the model, riboswitch data were not used for training. Instead, sequences with lengths within

128 nucleotides from the RNAstralign, TR0, and Rfam datasets were jointly used for training. The trained model predicted the affinity of 7107 sequences, and the Pearson correlation coefficient between the predicted affinity and the experimentally measured affinity was calculated for each dataset to evaluate the model.

DSRNAFold predicts the pairing probability of the end base of the constraint structure within the range of 0 to 1. Due to the simplicity of the riboswitch constraint structure, DSRNAFold frequently predicts the end base pairing probabilities as either 0 or 1. Consequently, the predicted affinity, which

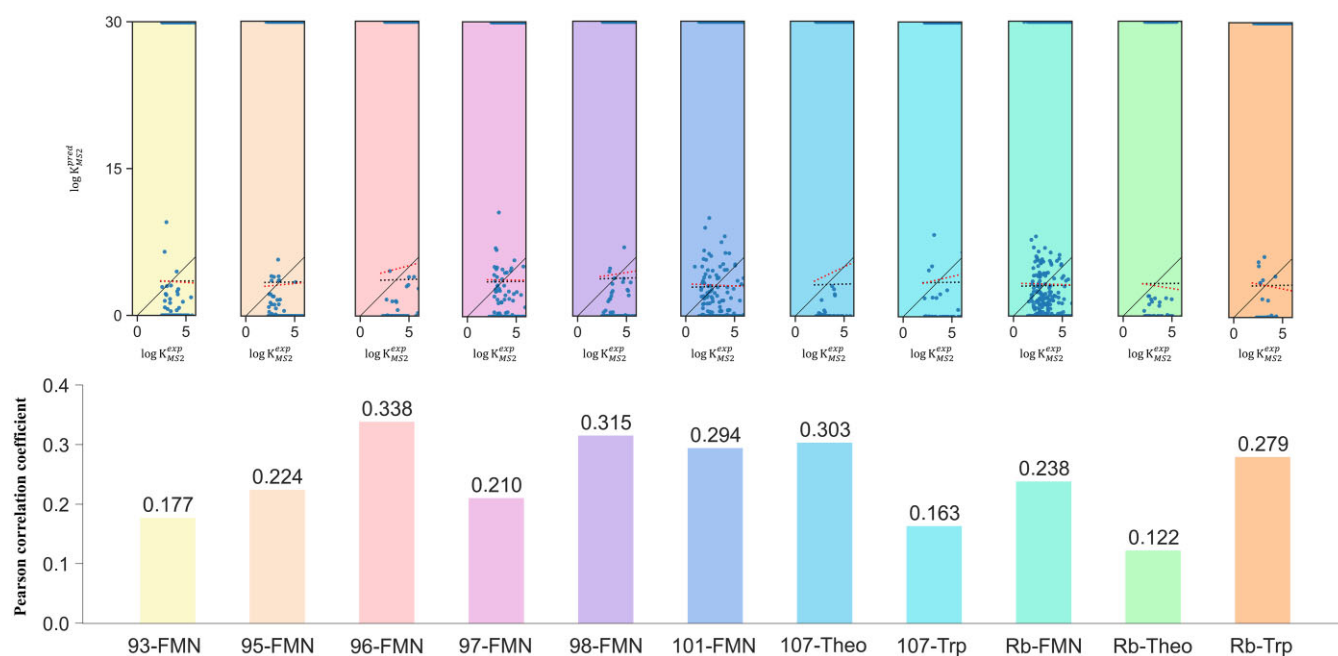


Figure 7. Correlation analysis of predicted and experimental binding affinities for various riboswitch datasets. The top scatter plots depict the relationship between the logarithm of experimentally measured binding affinities and the predicted values for each dataset, with fitted curves illustrating the influence of extreme values. The bottom bar chart presents the Pearson correlation coefficients, reflecting the strength of correlation between predicted and experimental affinities; higher coefficients indicate stronger correlations.

is calculated as the logarithm of the ratio of these probabilities, often reaches extreme values. In cases where the end base pairing probability is 0, the value is mapped to the top of the coordinate axis. Conversely, when the pairing probability is 1, the value is mapped to the bottom.

It is evident that there are numerous points concentrated at these extreme positions (Fig. 7). The black dotted line represents the fitting curve including these extreme values, while the red dotted line represents the fitting curve after removing the extreme values. The red dotted line, excluding the extreme values, fits the data better and is closer to the true value. The Pearson correlation coefficients for each dataset range from 0.122 to 0.338, with higher values indicating a stronger correlation between predicted and experimental binding affinities. The datasets 96-FMN and 98-FMN exhibit the highest correlation coefficients, 0.338 and 0.315, respectively, indicating higher prediction accuracy for these riboswitches. Other datasets like 107-Trp and Rb-Theo show lower correlation coefficients. Although these coefficients are not as high, they still demonstrate a certain level of correlation, affirming that the model's predictions are generally applicable and useful. Additionally, while energy-based secondary structure prediction methods have proven effective in predicting the binding affinity of riboswitches, DSRNAFold offer a complementary and increasingly powerful tool for advancing our understanding and prediction of RNA behavior (see [Supplementary Fig. S6](#)).

Overall, these findings validate the capability of DSRNAFold in predicting riboswitch binding affinities, showcasing its potential for broader applications in RNA-based research and therapeutic development.

Visualization

After quantitatively evaluating the performance of the RNA secondary structure prediction model, we visualized the sec-

ondary structures predicted by DSRNAFold to further elucidate the advantages of our model. To accomplish this, we converted the predicted pairing interactions into bpseq format, leading to graphical representations rendered by our custom-developed visualization software, VaRNAfold. This software is capable of generating vector images for high-resolution display of RNA secondary structures. Simultaneously, we selected four mainstream secondary structure prediction models, Ufold, MXfold2, SPOT-RNA, and CONTRAfold, using our visualization tool to obtain visual results.

Two examples were selected from the *srp_Brad.spec._CU234118* and *RNaseP_A. telluris* sequences due to their biological significance and functional roles. The *srp_Brad.spec._CU234118* is part of the SRP, which is essential for directing proteins to their proper cellular locations during synthesis. *RNaseP_A. telluris* is a component of RNaseP, a ribonucleoprotein responsible for catalyzing the maturation of tRNA molecules, a crucial step in protein translation. Their well-documented functions in key cellular processes make them ideal candidates for assessing the applicability of predictive models. Upon a detailed examination of the RNA secondary structure predictions (Fig. 8), it is evident that our model's predictions align closely with the verified ground truth, exhibiting our model's outstanding level of detail and accuracy. Hairpin loops, which are pivotal for RNA function due to their role in stabilization and molecular recognition, are rendered with exceptional precision. The size and location of these loops are almost identical to those of the ground truth, indicating a robust understanding of the RNA folding principles embedded in our model's algorithm. Furthermore, the internal loops and bulges, which can have significant implications on the dynamics and mechanical properties of the RNA molecule, are also predicted with a high degree of accuracy in DSRNAFold (Fig. 8A). These regions, despite being challenging due to their variable nature,

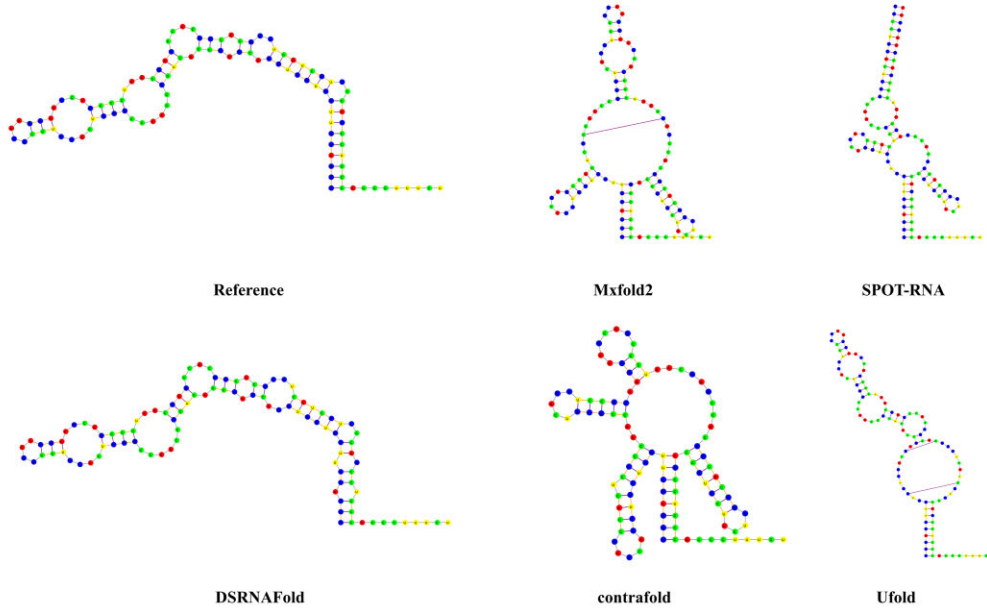
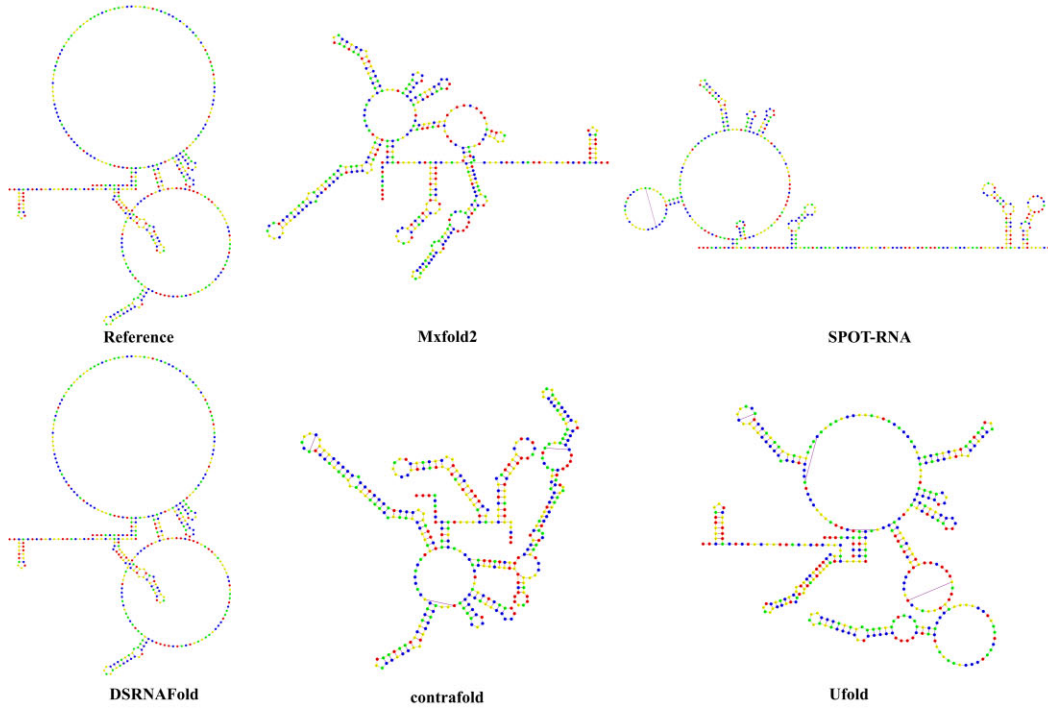
A *srp_Brad.spec._CU234118***B** *RNaseP_A.telluris*

Figure 8. Visual comparison of models for two experimentally determined RNA secondary structures. It representations show RNA secondary structures predicted by DSRNAFold compared to other models, based on sequences *srp_Brad.spec._CU234118* and *RNaseP_A. telluris* from the Archivell dataset. Each node represents a nucleotide, and edges indicate backbone connections or base pairings. Stem regions and base-pair interactions are visually highlighted.

are rendered reliably, signifying our model's advanced ability to interpret and predict RNA secondary structure nuances. In contrast, the other models yield minor structural discrepancies. For MXfold2 and CONTRAfold, for instance, slight deviations are observed in the arrangement of multistem junctions, which are critical for three-dimensional folding and hence the functionality of RNA. SPOT-RNA and Ufold

also contribute valuable insights, but they exhibit limitations in modeling more complex structures.

The RNA molecule *srp_Brad.spec._CU234118* has a structurally intricate architecture characterized by several expansive loop formations alongside numerous diminutive hairpin loops, as well as regions of unpaired nucleotides. The pronounced loop structures are attributed to long-range base

pairing interactions (Fig. 8B). The proposed model, DSRNAFold, adeptly identifies these principal loop configurations, indicating its notable proficiency in capturing the structural attributes associated with extensive base pairing. Furthermore, while the Ufold and SPOT-RNA models can approximate the overarching loop features, their delineation of more granular structural elements may benefit from further refinement. MXfold2 and CONTRAfold demonstrate notable accuracy in predicting hairpin loops, internal loops, and bulges, albeit with varying degrees of deviation. Such disparities may be indicative of the ability of these models to process the complexities associated with long-distance base pairing and intricate structural conformations.

Discussion

In this study, we aimed to improve RNA secondary structure predictions by addressing key challenges such as data scarcity and overfitting. Our novel approach, DSRNAFold, integrates RNA sequence and structural context information in a phased learning strategy, using pairing constraints to train models on folding scores. This methodology not only enhances the robustness of predictions but also effectively handles both local and long-range nucleotide interactions. Across multiple benchmark datasets, DSRNAFold demonstrated superior performance compared to existing methods, particularly in pseudoknot recognition and chemical mapping reactivity, aligning with our study's goal to provide a more accurate tool for RNA analysis.

Notably, DSRNAFold's application of transfer learning represents a methodological innovation that improves its predictive accuracy and efficiency. By harnessing preexisting knowledge, this approach accelerates the model training process and reveals novel insights into RNA structuring, which is particularly beneficial for studying RNA molecules with sparse data. This aspect is particularly promising for the exploration of RNA molecules with limited available data, potentially accelerating the discovery of novel RNA-based therapeutics.

Furthermore, the adept handling of chemical mapping data by DSRNAFold represents a significant advance in RNA structural biology, offering a detailed framework for exploring the dynamic landscapes of RNA molecules. This ability not only enriches our understanding of RNA functional diversity but also has profound implications for elucidating disease mechanisms, thereby bridging gaps between computational predictions and experimental validation. In addition to demonstrating the efficacy of DSRNAFold in predicting riboswitch binding affinities, it is essential to highlight the complementary strengths of energy-based and AI-based methods. While energy-based approaches leverage well-established thermodynamic principles to accurately model RNA folding and interactions, AI-based methods like DSRNAFold offer significant potential through their ability to capture complex, nonlinear relationships within large datasets. These AI models can generalize well to new sequences and integrate diverse types of data to enhance predictive accuracy. The combination of these methodologies can lead to more robust and comprehensive RNA structure prediction tools, providing valuable insights that drive forward RNA-based research and therapeutic development.

Despite the groundbreaking achievements of DSRNAFold, we recognize several challenges that impact its broader applicability and accuracy across diverse RNA structures. The

complexity of RNA molecules requires models that are precise yet capable of generalizing across a wide array of structures. Issues such as overfitting and computational efficiency are critical, as they significantly affect the model's real-world utility. However, these limitations do not undermine the study's results because our validations across multiple benchmarking datasets have demonstrated the robustness and reliability of DSRNAFold under varied conditions, but highlight areas for future improvement. For instance, integrating more diverse training datasets or large-scale embeddings [61] with structure-aware neural architectures could further enhance the model's performance and reliability.

The findings from this study have broad implications, extending beyond RNA biology to potentially impact fields such as personalized medicine and drug discovery. One promising avenue for future research is the integration of DSRNAFold with RNA sequence design technologies. By coupling our model with state-of-the-art sequence generation techniques, we could further enhance the precision of RNA constructs tailored for specific therapeutic applications. This integration has the potential to not only improve our understanding of RNA mechanisms but also streamline the development of RNA-based therapeutics. Additionally, expanding the model to accommodate other molecular structures could open new avenues in computational biology, providing a more holistic view of cellular functions. The ability of DSRNAFold to adapt to various molecular complexities, informed by insights from chemical mapping and sequence design, could lead to more effective and efficient predictive models. Such models not only would facilitate the exploration of RNA molecules with sparse data but could also revolutionize our approach to designing RNA sequences that are optimized for specific functions, thus impacting drug design and personalized medicine strategies.

In conclusion, DSRNAFold represents a significant step forward in achieving precise and efficient RNA secondary structure prediction. Compared to traditional RNA prediction methods that often rely on rigid, pre-established thermodynamic models, DSRNAFold employs a novel deep learning framework that autonomously identifies complex structural motifs. This approach not only enhances accuracy by reducing reliance on extensive empirical data but also increases prediction efficiency by leveraging state-of-the-art machine learning techniques such as transfer learning. This innovation allows DSRNAFold to adapt more dynamically to new data, making it highly effective even for RNA molecules with limited known data. Moreover, DSRNAFold's ability to integrate and interpret chemical mapping data demonstrates that it offers new avenues for understanding RNA functions and developing RNA-based therapeutics. Overall, DSRNAFold provides a more robust, adaptable, and accurate tool for RNA research, which could have far-reaching implications in fundamental biological research, the conceptualization of targeted therapeutics, and the advancement of personalized medicine.

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Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

All data sets analyzed in this article are publicly available. Results used for figures in this paper are provided in Supplementary Section S2. The original Cloud Lab datasets are available from the RNA Mapping Database [39] under accession IDs ETERNA_R00_0000 (round 0), ETERNA_R69_0000 (round 1), ETERNA_R70_0000 (round 2), ETERNA_R71_0000 (round 3), ETERNA_R72_0000 (round 4), ETERNA_R73_0000 (round 5), ETERNA_R74_0000 (round 6), ETERNA_R75_0000 (round 7), ETERNA_R76_0000 (round 8), ETERNA_R77_0002 (round 9), ETERNA_R78_0001 (round 10), ETERNA_R79_0001 (round 11), ETERNA_R80_0001 (round 12), ETERNA_R81_0001 (round 13), ETERNA_R82_0001 (round 14), ETERNA_R83_0003 (round 15), ETERNA_R84_0000 (round 16), ETERNA_R85_0000 (round 17), ETERNA_R86_0000 (round 18), ETERNA_R87_0001 (round 19), ETERNA_R89_0000 (round 20), ETERNA_R91_0000 (round 21), ETERNA_R92_0000 (round 22), and ETERNA_R94_0000 (round 23). The DSRNAFold source code used in this study and all analyzed datasets are available at Zenodo (<https://doi.org/10.5281/zenodo.15545891>). The DSRNAFold web server is available for use at <http://www.nextbioai.com/>.

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